Theorem 17: inapproximability of the optimal-transport operator p-distances

Proof overview of the Lean 4/Mathlib formalisation

Folder `GsimOT/Theorem18_OperatorNormHardness`

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport.*
File `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository; statement, equation, page and line numbers refer to this PDF. `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

1 Statement

Paper. Theorem 17 (Section 4, p. 12, l. 432–433; proof in App. G, pp. 54–57, l. 1710–1804): *for* $1 \le p \le \infty$ *and any* $\varepsilon > 0$*, unless* P = NP*, there is no polynomial-time approximation algorithm computing* d_p^{OT} *with multiplicative error guarantee* $2^{b(p)} - \varepsilon$, where $b(p) = \max\{1/p, 1 - 1/p\}$ and $b(\infty) = 1$ (l. 431). It gives Theorem 4 (2) (p. 4, l. 135–137); the restriction to maximum degree two for $p = 1$ is Theorem 16. The tools are the aggregated matrix $A^Q_{G,H} = A_G - QA_HQ^\top$ of Eq. (45) (p. 53), Lemma 53 (p. 53, $\|A^Q_{G,H}\|_p \le \|B_p(G,H,Q)\|_p$) and Corollary 54 (p. 54, a column with k entries of size $\ge \varepsilon$ gives $\|A^Q_{G,H}\|_p \ge k^{b(p)}\varepsilon$). The reduction from Three-Partition takes the paths and cycles of Theorem 16 and attaches M leaves to every vertex; YES-instances give $d_p^{\mathrm{OT}}(G,H) \le 1$, NO-instances $d_p^{\mathrm{OT}}(G,H) > 2^{b(p)} - \varepsilon^*$ (p. 54, l. 1728–1730).

Lean. Only the gap (the correctness of the reduction) is formalised. For an instance I, the graphs are $G =$ `I.graphG M` $=$ `addLeaves (pathUnion I.a) (fun _ => M)` and $H =$ `I.graphH M` $=$ `addLeaves (cycleUnion I.m I.A) (fun _ => M)`, and $1 \le p \le \infty$ (`p : ENNReal` with `Fact (1 <= p)`).

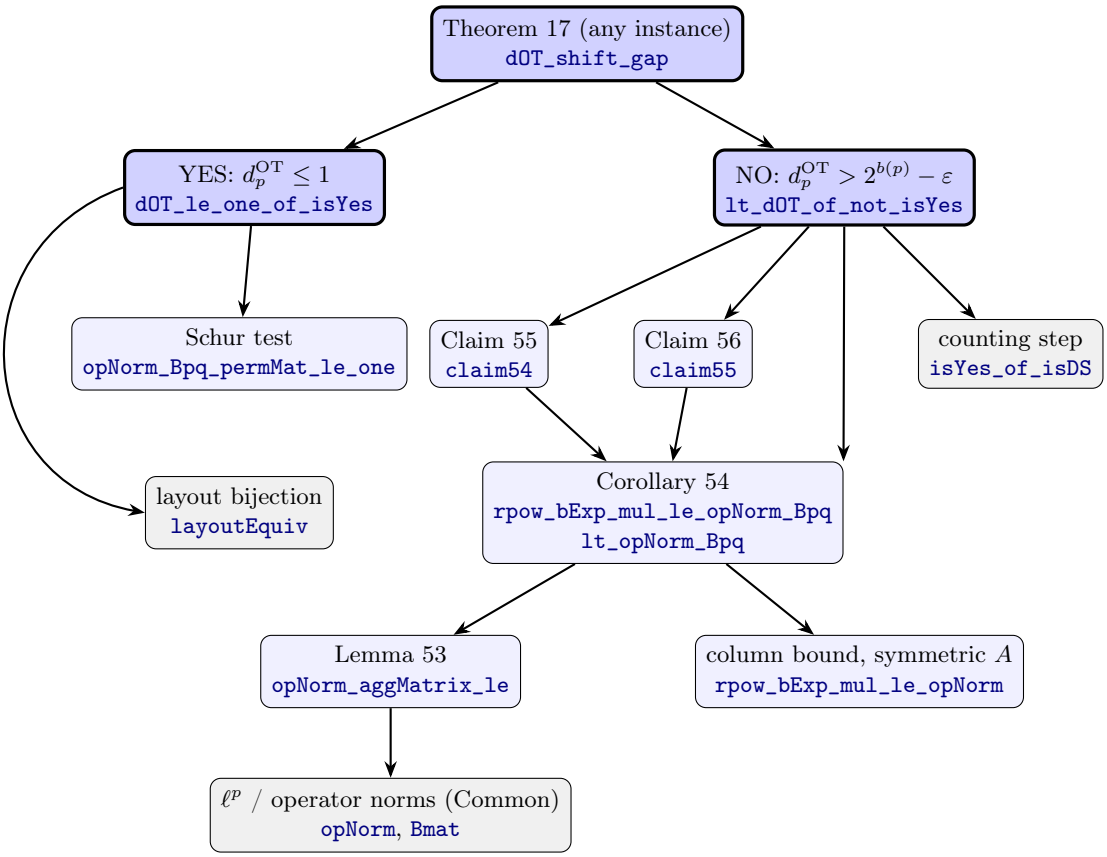
- `GsimOT.Hardness.ThreePartition.dOT_le_one_of_isYes`: if $a_i \ge 2$ for all i and I is a YES-instance, then $d_p^{\mathrm{OT}}(G,H) \le 1$ (for every M);
- `GsimOT.Hardness.ThreePartition.lt_dOT_of_not_isYes`: if $a_i \ge 3$ for all i, I is a NO-instance, $0 < \varepsilon \le 1$ and $M > \max\{2(4/\eta)^2 + 1,\ 2(2/\varepsilon')^2 + 1\}$ with $\eta = \varepsilon/8$, $\varepsilon' = (\varepsilon/8)/(mA - 1)$ (Eq. (46)), then $2^{b(p)} - \varepsilon < d_p^{\mathrm{OT}}(G,H)$;
- `GsimOT.Hardness.ThreePartition.dOT_shift_gap`: both, for every instance, after the w.l.o.g. shift (add 3 to each a_i and 9 to A; the paper assumes $a_i \ge 3$ w.l.o.g., l. 1712).

$d_p^{\mathrm{OT}} =$ `dOT p p` is the infimum over $\mathrm{DS}(V,W)$ of $\|B_p(G,H,Q)\|_p$ (`Bmat p p`, `opNorm p p`; `Common/OTDistances.lean`, `Common/OperatorNorm.lean`). The Lean statement is the paper's gap, for all $0 < \varepsilon \le 1$; this range contains the paper's $0 < \varepsilon^* < \sqrt{2} - 1$ (l. 1728).

2 Where the proof lives

File	Content
`OperatorPNorms.lean`	$b(p)$ (`bExp`), Jensen (`rpow_sum_mul_le`), Schur test (`opNorm_le_one_of_sum_abs_le_one`), the column bound of Corollary 54 for symmetric matrices (`rpow_bExp_mul_le_opNorm`).
`Lemma19_AggregatedMatrix.lean`	Eq. (45) (`aggMatrix`, `aggMatrix_isSymm`) and Lemma 53 (`opNorm_aggMatrix_le`).
`Corollary20_ColumnBound.lean`	Corollary 54 (`rpow_bExp_mul_le_opNorm_Bpq`, `lt_opNorm_Bpq`).
`Claim54_LeafMass.lean`	the observation on A^Q for (non-)edges (`aggMatrix_of_adj`, `aggMatrix_of_not_adj`), $\lvert N_G(v)\rvert \ge M + 1$, Claim 55 (`claim54`).
`Claim55_SplitMass.lean`	Claim 56 (`claim55`).
`Theorem18.lean`	the graphs G, H, the YES-direction, the final argument (NO-direction), the shifted version.
`Common/ThreePartition.lean`, `Common/PathsAndCycles.lean`, `Common/Leaves.lean`	instances, layout bijection, counting step `isYes_of_isDS`, paths, cycles, leaves.

3 Proof layout



Blue: this folder; grey: `GsimOT/Common`. Nothing is cited.

4 The proof step by step

Eq. (45) and Lemma 53 (pp. 52–54, l. 1671–1697). $A^Q_{G,H}(v,v') = \sum_{w,w'} Q(v,w)D(v,w,v',w')Q(v',w')$ is `aggMatrix`; it is symmetric (`aggMatrix_isSymm`, used in Corollary 54). Lemma 53 is `opNorm_aggMatrix_le`. The paper picks optimal test vectors x, y with Lemma 19 (duality, App. A, p. 14) and lifts them to $\tilde{x}(v,w) = Q(v,w)^{1/p}x(v)$, $\tilde{y}(v,w) = Q(v,w)^{1-1/p}y(v)$, so that $y^\top A^Q x = \tilde{y}^\top B_p \tilde{x}$. The formalisation uses the same $\tilde{x}$ but no $\tilde{y}$: it writes $(A^Q x)(v) = \sum_w Q(v,w)^{1-1/p}(B_p\tilde{x})(v,w)$ and bounds $\|A^Q x\|_p \le \|B_p\tilde{x}\|_p$ by Jensen (`rpow_sum_mul_le`), since the rows of Q sum to 1. The convention $0^0 = 0$ is `powZ`.

Corollary 54 (p. 54, l. 1699–1709). If column v of A^Q has k entries of absolute value $\ge \varepsilon$, then $\|A^Q\|_p \ge k^{b(p)}\varepsilon$ (`rpow_bExp_mul_le_opNorm`). The bound $k^{1/p}\varepsilon$ comes from e_v, as in the paper; for $k^{1-1/p}\varepsilon$ the paper uses $\|A\|_p = \|A^\top\|_{p^*}$ and symmetry, the formalisation tests A directly with a normalised sign vector on the k rows, which is the same computation. With Lemma 53 this gives `rpow_bExp_mul_le_opNorm_Bpq`, and `lt_opNorm_Bpq` is the "in particular" $(k > (c/\varepsilon)^{1/b(p)} \Rightarrow \|B_p\|_p > c)$.

Construction (p. 54, l. 1710–1727). G, H are `graphG`, `graphH` (`addLeaves`); base vertices are `Sum.inl`, leaves `Sum.inr`. The w.l.o.g. $a_i \ge 3$ is `ThreePartition.shift`; $A \ge 7$ is derived in `lt_dOT_of_not_isYes` (`hA`). The observation $A^Q(x,x') = \sum_{yy' \notin E(H)} Q(x,y)Q(x',y')$ for $xx' \in E(G)$ is `aggMatrix_of_adj`; its counterpart for non-edges is `aggMatrix_of_not_adj`.

YES-instances (p. 55, l. 1733–1737). The layout bijection τ of three paths per cycle (`layoutEquiv`, `layout_adj_cycleUnion`) is extended to the leaves by `Equiv.sumCongr` and `Equiv.sigmaCongrLeft`. The paper observes that $A_H - A_G^\pi$ is the adjacency matrix of $3m$ disjoint edges, so its norm is 1. The formalisation shows that every vertex has at most one mismatch and bounds $\|B_p(G,H,P_\pi)\|_p \le 1$ by the Schur test (`opNorm_Bpq_permMat_le_one`, `opNorm_le_one_of_sum_abs_le_one`).

Claim 55 (p. 55, l. 1741–1759). `claim54`: if $Q(v, L_H) \geq \eta$ and $M > 2(4/\eta)^2 + 1$, then $\|B_p\|_p > 2^{b(p)}$. The case split "for all w, $Q(v, L(w)) \leq \eta/2$" (`hcase`) and the two lower bounds $\eta/2$ (Case 1) and $\frac{\eta}{2}(1-\delta)$ with $\delta = 1/2$ (Case 2) follow the paper line by line; the at most $\lfloor 1/\delta \rfloor = 2$ heavy vertices are counted by `card_mul_le_of_lt_mass`. The paper's size condition $M - 1 > 2(4/\eta)^2$ is matched to Corollary 54 by `rpow_inv_bExp_le`: $(2^{b(p)}a)^{1/b(p)} \leq 2a^2$ for $a \geq 1$.

Claim 56 (pp. 55–56, l. 1760–1773). `claim55`: if $Q(v, w), Q(v, w') \geq \varepsilon$ for distinct base vertices with at most one common neighbour and $M > 2(2/\varepsilon)^2 + 1$, then $\|B_p\|_p > 2^{b(p)}$. The paper's "$|I| \leq 1$ since $|V(C_j)| = A \geq 5$" (l. 1762) is the hypothesis `hI`, discharged by `card_common_adj_cycleUnion_le_one`.

Final argument (pp. 56–57, l. 1774–1804). In `lt_dOT_of_not_isYes`: if Claim 55 or Claim 56 applies, done (`hA'`, `hB`). Otherwise every base vertex v sends mass $\geq 1 - \varepsilon/8 - \varepsilon/8 = 1 - \varepsilon/4$ to a single base vertex $f(v)$ (`hf`, the paper's $Q(v, w) \geq 1 - \varepsilon - \eta$); since $1 - \varepsilon/4 > 1/2$, f is injective, hence the bijection τ (`hf_inj`, `Equiv.ofBijective`). As I is a NO-instance, some edge vv' of a path is mapped into two different cycles (`hcross`, via the counting step `isYes_of_isDS`); with $a_i \geq 3$ the vertex v has a second neighbour v'' (`exists_other_adj`). Either $\tau(v'') \not\sim \tau(v)$ (an entry $\geq (1 - \varepsilon/4)^2$ by `hentry_adj`) or $\tau(v)$ has a cycle neighbour $w^* \neq \tau(v'')$ whose preimage is not adjacent to v (an entry $\leq -(1 - \varepsilon/4)^2$, via `aggMatrix_of_not_adj`). Two entries in column v and Corollary 54 with $k = 2$ give $\|B_p\|_p \geq 2^{b(p)}(1-\varepsilon/4)^2 > 2^{b(p)} - \varepsilon$ (`hL`, using $2^{b(p)} \leq 2$). The infimum over Q is handled by `le_iInf_isDS`.

5 Assumptions and trusted base

No cited results and no hypotheses beyond those of the statement. Only the standard axioms `propext`, `Classical.choice`, `Quot.sound`. The complexity-theoretic part (polynomial size of the reduction, NP-hardness of Three-Partition) is not formalised.

6 Notes on the formalisation

- Lemma 53 is proved by Jensen instead of the duality of Lemma 19; Corollary 54 uses a sign vector instead of $\|A\|_p = \|A^\top\|_{p^*}$ (see above). The bounds are the same.
- YES-direction: the Schur test (bound ≤ 1) instead of the computation of the norm ($= 1$); `dOT_le_one_of_isYes` assumes $a_i \geq 2$.
- ε ranges over $(0, 1]$, which contains the paper's range $(0, \sqrt{2} - 1)$ for ε^* (l. 1728); the paper uses $\varepsilon^* < \sqrt{2} - 1$ to conclude $2^{b(p)} - \varepsilon^* > 1$ (l. 1729–1730).
- Remark 62 (other operator $p \to q$-norms, p. 62, l. 1989–1998) is not formalised.

7 How to check this result

- `lake build GsimOT.Theorem18_OperatorNormHardness.Theorem18`
- `lake env lean scripts/CheckAxioms.lean` prints the axioms of `GsimOT.Hardness.ThreePartition.dOT_le_one_` and `GsimOT.Hardness.ThreePartition.lt_dOT_of_not_isYes`.
- The table `Theorem18_correspondence.pdf` in this folder lists every paper item with its Lean declaration and line.